AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 13 · STUDENT LESSON

Train a Classifier

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Fit once. Keep the state. Reproduce the whole run.

Build and inspect a consistent training-only scaled-neighbour pipeline.

Keep one habit: Complete training contract → runnable fit/predict script → unit-conversion contrast → reproducibility probes → independent new pipeline.

MAIN LESSON · SECTION 1

Fit once and apply the same pipeline to every raw query

> Reproducing the same output does not establish that the output is right.

Learning goals

I can connect **training**-only scaling and neighbour **prediction** in one **pipeline**.

I can **reproduce predictions** from recorded data settings and fitted values.

I can audit a **pipeline** for re**fitting** feature-order errors and evaluation leakage.

Recall the fitted scale and baseline

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.
- Which values does min-max scaling learn, and which values should be reused for a validation query?

▼ **Optional readiness · Candidate rule search**

How does a **training**-majority baseline choose its **prediction**, and where is its validation accuracy measured?

> **My readiness explanation**

End goal

By the end, I can explain train a classifier using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Training fits model **parameter**s from labeled examples; **prediction** applies the frozen fitted model to new feature rows.



LEARNING TARGET **Training**: spots 0:H,2:H,4:S,6:S. Compare rules S if spots>=1,>=3,>=5. Choose by **training** accuracy, then **predict** held-out 1:H and 5:S without revising. Report both stages.

Builder name and experiment date:

Readiness for the connected explanation

Readiness for Searching candidate rules

This week's end goal

MAIN LESSON · SECTION 2

Read feature order, units and source roles before fitting

> A raw pair only has meaning under its stated contract.

Pipeline idea and fictional source

A **pipeline** connects steps so a new query receives the same preparation as the **training** examples. This week combines the **training**-only scale introduced in Week 8 with a one-neighbour classifier. We will fit once, keep the **fitted state**, and call one **prediction** function for every raw query.

Use the same fictional trip data. Each feature pair is ordered (distance in cm, brightness in sensor units). **Training** IDs are A,B,C.

Validation IDs V1,V2,V3 belong to separate trips; related trips were grouped before splitting. Use min-max scaling, sum of absolute scaled differences, k=1 and alphabetical ID order for exact distance ties.

Complete three-row training table

Training ID	Feature pair	Label
A	(10,200)	clear

Training ID	Feature pair	Label
B	(20,100)	blocked
C	(20,200)	blocked

Input and validation assumptions

The **input contract** requires two finite numeric features in the stated order and units, with no missing values. **Training** also needs unique IDs, known labels, at least one row and nonzero ranges.

These assumptions must be checked before this short classroom program is used; its explicit guard only checks zero ranges.

▼ Optional extension · Separate cutoff-search source and contract

The majority baseline from Week 10 ignored scene features. This week a small **fitting** algorithm will choose a rule that uses one feature. We supply the rule form: **predict** blocked when a distance reading is at or below a cutoff, otherwise **predict** clear. The algorithm will try three cutoffs and keep the one making the fewest **training** mistakes.

The teacher-authored fictional rows contain a sensor distance in centimetres and a separate reviewed lane label, blocked or clear. The reading is available before **prediction**; the label comes from the scene review.

A single distance need not describe every obstruction. These numbers are for learning, not real robot control. Each row comes from a different trip, and **training** and validation trips are separate.

Training row	Distance in cm	Reference label
1	2	blocked
2	4	blocked
3	6	clear

Training row	Distance in cm	Reference label
4	8	clear

Before scoring, declare cutoffs 3, 5 and 7 cm in ascending order, **predict** blocked at equality, and keep the smaller cutoff if **training error**s tie. This search tests only these three rules with this direction.

It does not find the best possible model or prove that distance causes the label.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use train a classifier to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Apply the connected idea: Searching candidate rules

MAIN LESSON · SECTION 3

Save the training scale and stored examples as one fitted state

> Prediction transforms a raw query without learning a new scale.

Complete fit/scale/predict script

```
def scale(x, low, span):
    return tuple((x[j] - low[j]) / span[j] for j in range(2))

def fit(rows):
    xs = [x for item_id, x, label in rows]
    low = tuple(min(x[j] for x in xs) for j in range(2))
    span = tuple(max(x[j] for x in xs) - low[j]
                 for j in range(2))
    if 0 in span:
        raise ValueError("Review a constant training feature")
    stored = tuple((i, scale(x, low, span), y) for i, x, y in rows)
    return low, span, stored

def predict(x, model):
    low, span, stored = model
    q = scale(x, low, span)
    ranked = sorted((sum(abs(a - b) for a, b in zip(q, v)), i, y)
                    for i, v, y in stored)
    return ranked[0][1], ranked[0][2]

train = [("A", (10, 200), "clear"),
         ("B", (20, 100), "blocked"),
         ("C", (20, 200), "blocked")]
model = fit(train)
queries = [(11, 120), (19, 190), (12, 80)]
for x in queries:
    print(predict(x, model))
```

Trace stored state and the three outputs

fit returns minima (10,100), ranges (10,100) and scaled **training** examples. **predict** transforms a raw query with that state, ranks distances, then returns the selected ID and label. zip pairs matching features; sum adds their absolute differences. sorted uses distance first and unique ID second.

Run the whole script in Python 3. It prints ('A', 'clear'), ('C', 'blocked'), ('B', 'blocked') on separate lines. No validation labels enter fit or **predict**.

This is a small **pipeline** written with functions, not the scikit-learn **Pipeline** class.

▼ Optional extension · Complete candidate-cutoff search

```
train = [(2, "blocked"), (4, "blocked"),
         (6, "clear"), (8, "clear")]
best_errors = len(train) + 1
best_cutoff = None
for cutoff in [3, 5, 7]:
    errors = 0
    for distance, label in train:
        guess = "clear"
        if distance <= cutoff:
            guess = "blocked"
        if guess != label:
            errors = errors + 1
    print(cutoff, errors)
    if errors < best_errors:
        best_errors = errors
        best_cutoff = cutoff
print("chosen", best_cutoff)
```

The **output** lines are 3 1, 5 0, 7 1, then chosen 5. Cutoff 3 misses the blocked row at 4 cm. Cutoff 5 matches all four **training** labels. Cutoff 7 wrongly marks the 6 cm clear row as blocked.

Fitting stores cutoff 5; **prediction** then applies that stored rule to a new distance.

Separate search from later scoring

STAGE	EVIDENCE	DECISION
Try cutoffs 3 5 7	Training labels	Count 1 0 1 errors
Store cutoff 5	Smallest training error	Freeze this rule
Compare with baseline	Same validation rows	Report both scores

The supplied search chooses a cutoff using training errors. A perfect training fit does not establish performance on new examples.

best_errors starts above any possible mistake count. The strict < keeps the first tied candidate; ascending order makes that the smaller cutoff.

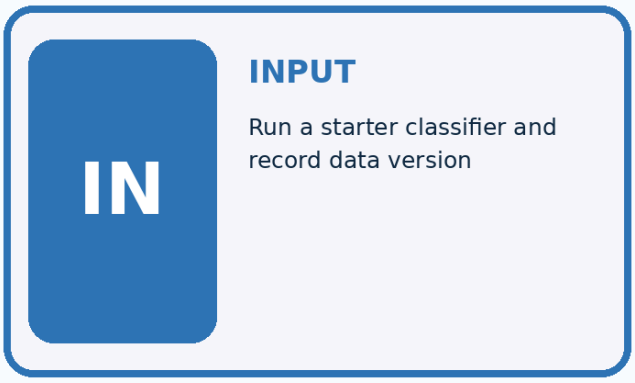
The code assumes nonempty **training** data, numeric distances and the two allowed labels. Validate those assumptions before using another dataset.

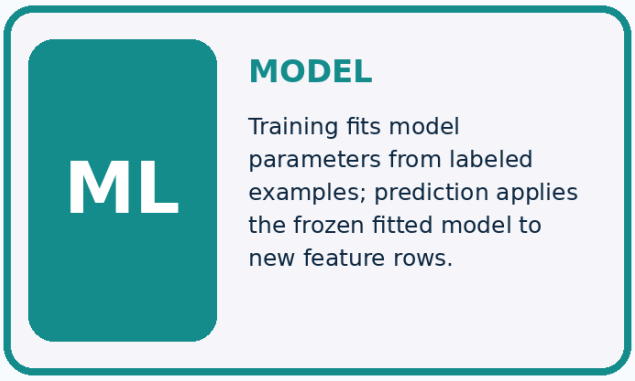
▼ **Earlier wording, alternatives and picture transcripts**

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Run a starter classifier and record data version, split, settings, and **prediction**s.

MODEL IDEA **Training** fits model **parameter**s from labeled examples; **prediction** applies the frozen fitted model to new feature rows.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Load data version v1 and a saved train/test split. Fit pre**process**ing on **training** rows only, then fit the classifier.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Apply the connected idea: Searching candidate rules

Connect the picture and the explanation

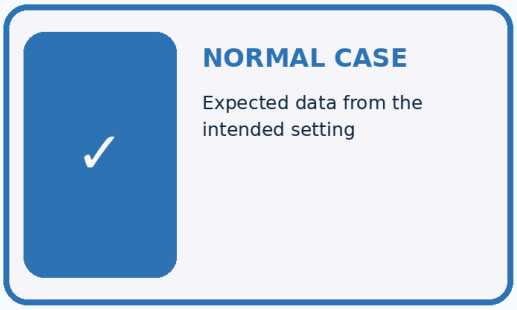
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

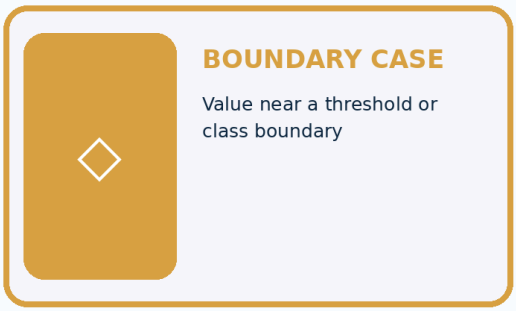
Compare reproducibility with a correct information boundary

> An unchanged mistake can reproduce perfectly.

Four evaluation conditions



NORMAL Expected data and expected behavior.

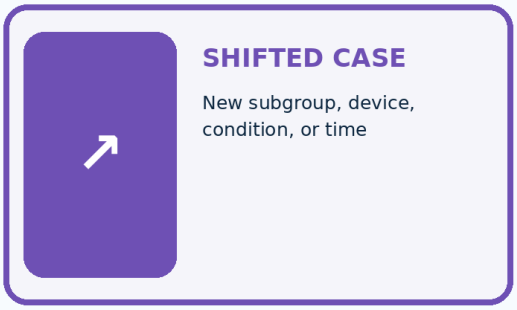


Keep this boundary clear

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Transform **training** but leave query numerically at 5.

▼ **Check after predicting**

Prediction: Distances become 15 and 55, incorrectly selecting healthy.

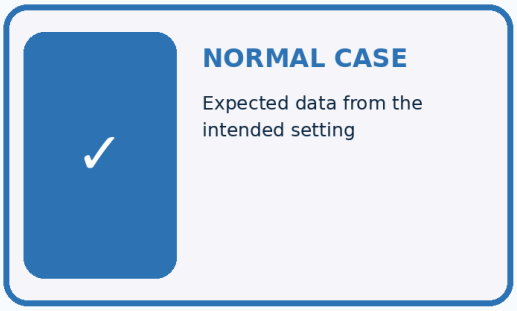
Reason: The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

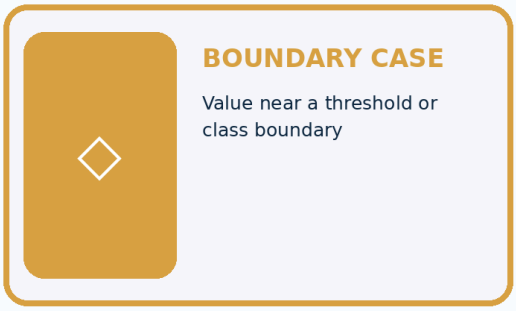
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

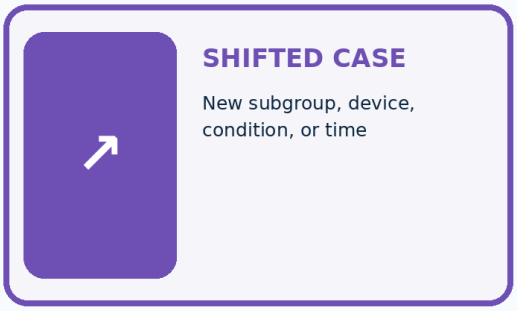
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace consistent preparation before nearest-neighbour prediction

> A saved scale, feature order and raw-input contract belong together.

Separate unit-conversion facts and scope

Separate a reproducible fitted **pipeline** from merely displaying **prediction**s.

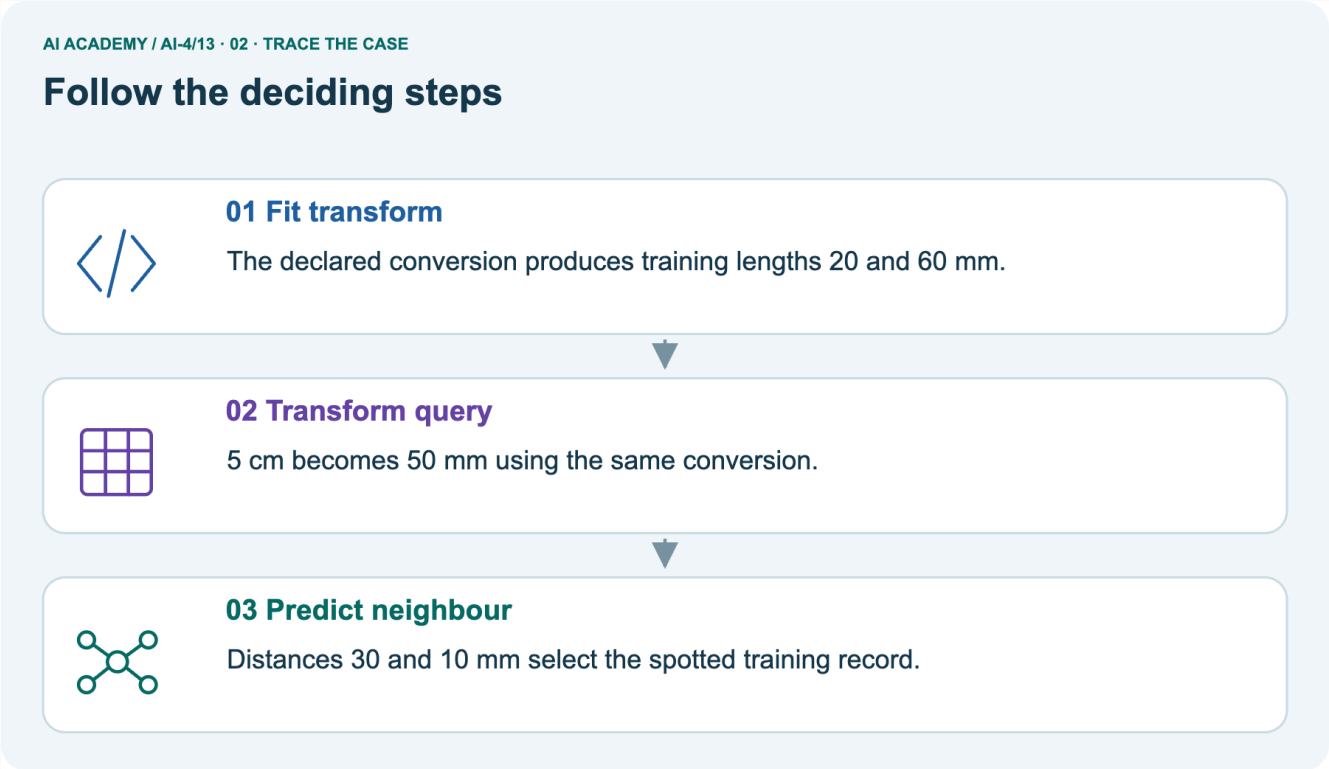
Toy **training** lengths are [2, 6] cm with labels [healthy, spotted]. **Pipeline** converts cm to mm, then uses k=1 absolute distance. Query is 5 cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The main pipeline learns minima and ranges from training. The pictured cm-to-mm conversion is different: multiplying by10 is a declared unit conversion, not a parameter learned from examples. Training lengths2/6cm become20/60mm; query5cm becomes50mm and selects the spotted record at distance10mm. Comparing the unconverted numeric5with20/60mixes units and incorrectly changes the neighbour.

Record whether each preparation step is a fixed rule or learned state; both must be applied consistently.

Trace the consistent pipeline



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit transform	The declared conversion produces training lengths 20 and 60 mm.
2. Transform query	5 cm becomes 50 mm using the same conversion.
3. Predict neighbour	Distances 30 and 10 mm select the spotted training record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

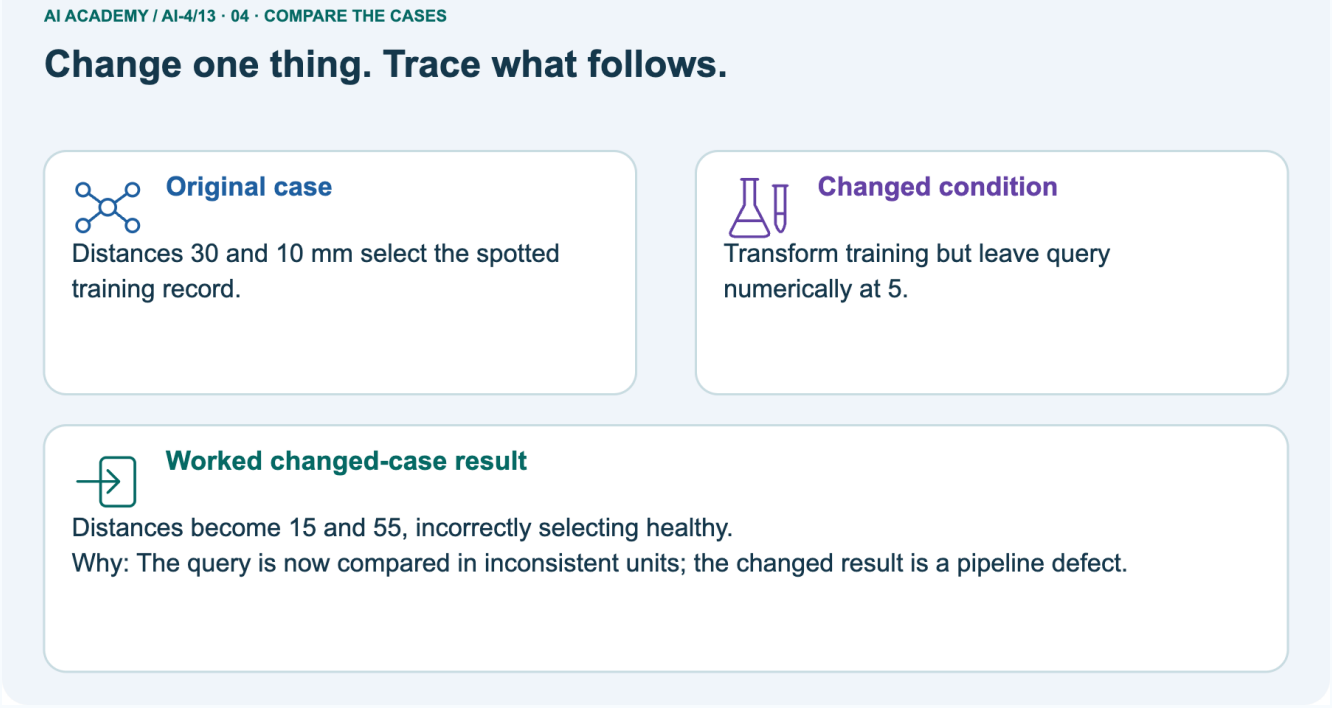
Read the picture in order. Distances 30 and 10 mm select the spotted **training** record. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict an unconverted query

- Change one thing: Transform **training** but leave query numerically at 5. **Predict** the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the evaluation and reproducibility limits

- A reproducible toy **pipeline** can still perform poorly; evaluation must use separate records and appropriate group splits.

- Read the labelled picture: Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main learned-scale query and swapped-order defect

Two stages with one fitted state

STAGE	INPUT AND ACTION	RESULT
Fit once	Training rows only	Scale plus stored rows
Predict each query	Raw pair plus state	Neighbour ID and label
Evaluate afterward	Prediction plus reference	Recorded match or error

Reference labels are used after prediction to score results. They do not enter the fitted scale or stored neighbour pool.

For raw query (11,120), scale gives (0.1,0.2). Distances to A,B,C are 0.9,1.1,1.7, so the function returns A and clear. It does not recalculate minima or add the query to stored rows.

The returned ID is useful for checking the reasoning, but it is not **evidence** that the label is true.

> **Trace (11,120) through both functions. What does fit store, what does predict reuse, and at which point may a validation reference label be compared?**

Store the **fitted state** together with the **input contract** and method settings. A model without its scale or feature order is incomplete.

Swapping the raw pair to (120,11) produces B and blocked here; the function cannot detect that the caller swapped the meanings.

▼ Optional cutoff validation comparison

Now reveal four validation rows in order: (3, blocked), (5, clear), (7, clear), (9, blocked). With the stored cutoff 5, **prediction**s are blocked, blocked, clear, clear.

Rows 1 and 3 match, so validation accuracy is 2/4 or 50 percent despite zero **training** mistakes.

The **training** labels tie at two blocked and two clear. Carry forward Week 10's baseline tie rule, choose clear. That baseline matches validation rows 2 and 3, also 2/4 or 50 percent. The fitted rule has not improved total accuracy on this set, and the two methods make different mistakes.

> **For the validation distance exactly 5 cm, what does the fitted rule predict and why? Which rows are wrong for the rule and the baseline? State their difference in percentage points.**

Keep the rule and baseline fixed for this comparison. Do not secretly choose a different cutoff after seeing validation labels and call that a **training**-only choice.

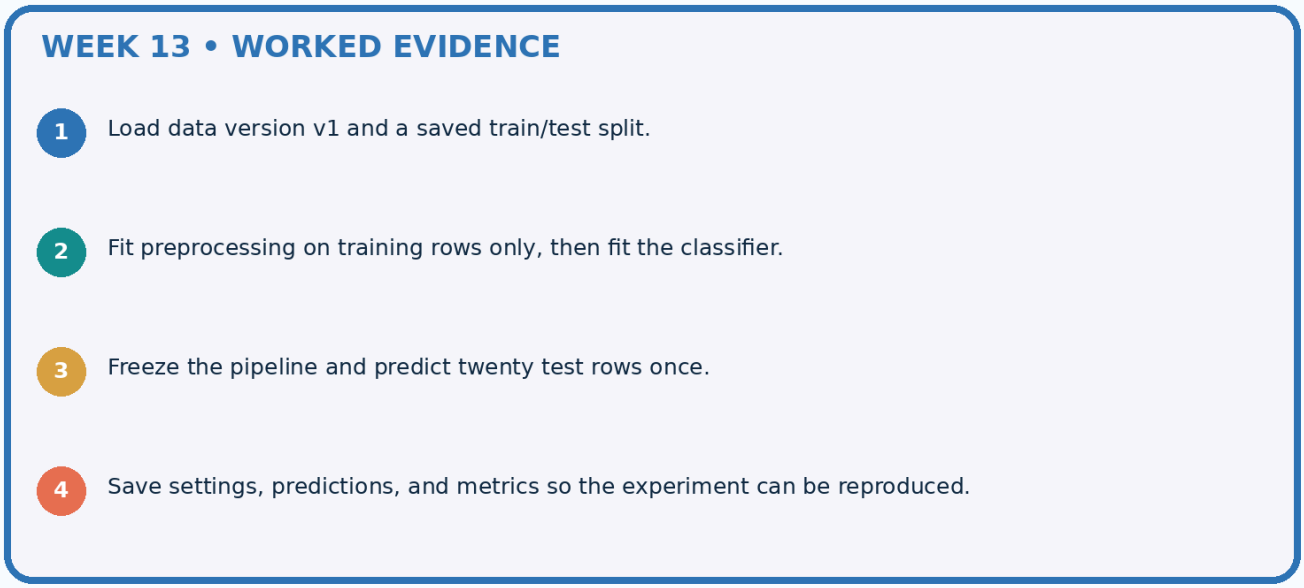
Validation can guide later development, but record those choices and reserve an untouched final test. No final test is supplied here.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



STEP 1 Load data version v1 and a saved train/test split.

STEP 2 Fit pre**process**ing on **training** rows only, then fit the classifier.

STEP 3 Freeze the **pipeline** and **predict** twenty test rows once.

STEP 4 Save settings, **prediction**s, and metrics so the experiment can be **reproduce**d.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Apply the connected idea: Searching candidate rules

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Train a Classifier

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a pipeline record mistaken for a correctness guarantee

> Check the source split, feature meaning and fitted state.

Main diagnosis and limits

- Check that fit receives **training** rows only, **predict** receives raw features in the correct order, and the **fitted state** stays unchanged during **prediction**.

  Do not scale a query twice, fit again on validation, or add labelled validation rows to the neighbour pool. A **pipeline** still leaks if the wrong rows are passed to its fit step.

- The tuple state groups the learned numbers and stored examples. Keep the associated data and code versions with it, and compare stored values and **output**s when repeating a run.

  Do not accept a matching score alone: two different **prediction** lists can have the same score.

> **A learner gets the same 2/3 score twice after fitting on training plus validation rows. They claim the pipeline must be valid because the result repeats. Explain why the claim fails and describe the repair.**

Challenge the shortcut

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE The model keeps learning whenever it sees test examples during evaluation.

▼ Optional cutoff-search and training-score limits

Humans chose the feature, labels, candidate cutoffs, direction and scoring method. The program used **training** examples to select the cutoff.

That is a small example of **fitting** a model; writing a loop alone does not make a method learn. Choosing the cutoff in advance without consulting **training** examples would instead be a fixed rule.

Record the source and split, rule form, candidate list, all **training error** counts, tie policy, selected cutoff and both validation **prediction** lists. Do not discard a difficult row to improve a score. The decision-tree lesson in Week 11 also used feature-based decisions; next week examines scores and calibration intuition.

> **A learner reports that cutoff 5 is perfect because training error is zero. Rewrite the claim using the training result, validation comparison and the limits of this small search.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The model keeps learning whenever it sees test examples during evaluation. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

Apply the connected idea: Searching candidate rules

MAIN LESSON · SECTION 8

Practise the saved run with less help

> A correct reproduction includes the documented V3 error.

Try the supported pipeline W2

Use workbook W2 for the connected example "A reproducible classifier **pipeline**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Record every fitted value, setting and prediction

Record dataset and code versions, **training** IDs A B C, validation IDs V1 V2 V3, feature order and units, fitted minima and ranges, k=1, distance and tie policy, zero-range policy and the actual Python version.

Save the **prediction**s in validation ID order. This script has no random step; a seed would not fix changed data or swapped features.

References for V1,V2,V3 are clear,blocked,clear. The printed **prediction**s clear,blocked,blocked match V1 and V2: 2/3. The **training**-majority blocked baseline matches only V2: 1/3.

Rerunning the same code and **input**s should **reproduce** these **output**s, including the V3 mistake. Reproducibility does not establish correctness or performance on new data.

Changing the scaling, k or feature list creates a new development run to record. Keep a separate final test for the final evaluation. Week 15 will use a confusion matrix to show label-error types; next week examines scores and calibration intuition.

Challenge automatic-leak-prevention reasoning

- Use the worked case for A reproducible classifier **pipeline**. Proposed explanation: "A **pipeline** prevents every leak automatically". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture's reproduction record

- Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve the new P/Q/R pipeline task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

<details>
<summary>Optional search-boundary and search-tie comparisons</summary>

A learner uses distance < cutoff instead of distance <= cutoff during **prediction**. That changes the boundary: at 5 cm the stored cutoff now **predict**s clear. Recheck code against the declared rule before interpreting the score.

A small symbol can change which examples match.

Another learner uses <= when deciding whether a candidate has fewer errors. That replaces an earlier candidate on a tie. The **prediction** boundary <= and the search comparison < have different jobs.

Use the declared tie policy; a tied **training** score gives no **evidence** that one tied rule will be better on new cases.

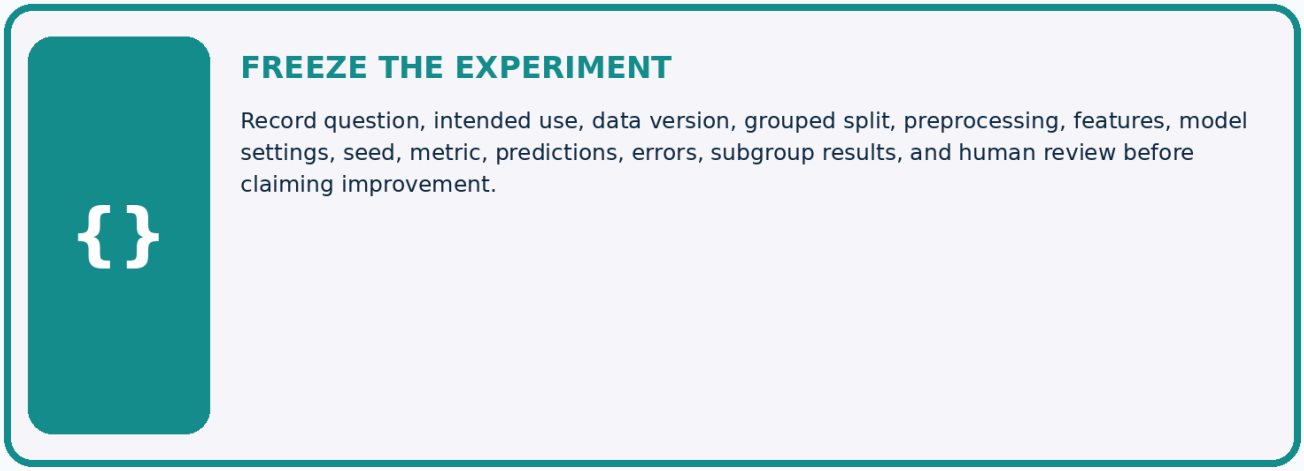
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: Zero **training error** means perfect **prediction**s
</details>

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be **reproduce**d from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success **threshold**

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to **reproduce** today's result:

A closer explanation

Apply the connected idea: Searching candidate rules

Practise with less help: Trace and score the saved run

Check your reasoning

Practise the connected topic: Searching candidate rules

Explain the picture

MAIN LESSON · SECTION 9

Run the complete reproducible classifier fixture

> Every query, reference and expected output is supplied and labelled.

Complete V1–V3 comparison table

Main validation ID	Raw distance/brightness pair	Reference	Printed nearest ID / prediction
V1	(11,120)	clear	A / clear
V2	(19,190)	blocked	C / blocked
V3	(12,80)	clear	B / blocked

Normal, tie, missing, outside-range, order and constant probes

Complete pipeline probe	Expected boundary or result
Normal raw(11,120)	Stored training scale gives(0.1,0.2), nearest A clear.

Complete pipeline probe	Expected boundary or result
Exact distance tie raw(10,100)	Scaled(0,0): A and B distance1,C2; alphabetical ID chooses A clear.
Missing query coordinate(11,None)	Original numeric contract is violated; request correction/review before arithmetic, not zero.
Outside training range raw(12,80)	Scaled(0.2,-0.2), nearest B blocked; no refitting or silent clipping.
Wrong feature order raw(120,11)	The unguarded function returns B blocked but cannot detect swapped meanings; repair the input contract.
Constant-feature training: A(10,100)clear,B(10,200)blocked	First training range0; original guard raises ValueError for review, before prediction.

Predict, fit, run and record

1. Starter Classifier Run: predict the complete script outputs, fit training A/B/C once, and run V1–V3 in their stated ID order. Compare actual outputs with the supplied trace and a peer's record; explain the first difference rather than hide it. Record training minima/ranges, scaled stored rows, feature order/units, k, distance/tie policy, code/data/Python versions and nonrandom status.

   Score predictions only afterward: pipeline2/3 versus training-majority blocked baseline1/3. Retain the V3 mistake in an unchanged rerun. Separate the optional cutoff-search dataset from the pipeline. Keep new P/Q/R rows for the independent task.

Keep evidence and actual execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Shifted
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Starter Classifier Run: trace code, predict outputs, execute the fixed pipeline, and compare recorded results with a peer.

1. Write the hypothesis and expected **evidence** before running.

2. Starter Classifier Run: trace code, **predict output**s, execute the fixed **pipeline**, and compare recorded results with a peer.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: fit and reproduce the new P/Q/R run

> Retain the incorrect query result rather than adjusting state to its reference.

Complete independent dataset and scope

Use P (0,60) blocked, Q (20,20) clear, R (20,60) clear as new **training** rows. Fit the same **pipeline**. **Predict** validation queries (2,28),(18,56),(4,12) with references blocked,clear,blocked.

Give fitted minima and ranges, transformed queries, selected IDs, **prediction**s, accuracy and **training**-majority baseline.

This new task uses training P/Q/R and its own reference labels, not A/B/C. Fit only those supplied training rows, use raw feature pairs in the same distance/brightness order, and keep the resulting state for all three queries. Reusing the earlier fitted A/B/C state would answer a different task. No final test or broad reliability evidence is supplied.

First attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Build and trace a fresh fitted state

Week 13 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A classifier fits on rows 1–80 and is evaluated on those same rows. Its accuracy is 100%. What additional **evidence** is needed before claiming generalization?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. **Predict** the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Train Plant Classifier v1 without opening the final test labels during model choice; save the experiment record.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Training: spots 0:H,2:H,4:S,6:S. Compare rules S if spots>=1,>=3,>=5. Choose by **training** accuracy, then **predict** held-out 1:H and 5:S without revising. Report both stages.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep a model record that includes its preparation

> A score and k alone cannot reproduce this classifier.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Train Plant Classifier v1 without opening the final test labels during model choice; save the experiment record.

Save the raw feature contract, dataset/group split version, code version, fitted minima/ranges/stored examples and method settings with the experiment. Keep ID-aligned references and predictions and the baseline’s denominator. Label a run traced, executed or not run honestly.

A fixed unit conversion differs from a learned transformation; neither permits fitting on validation or using final-test labels for model choice.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits, uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

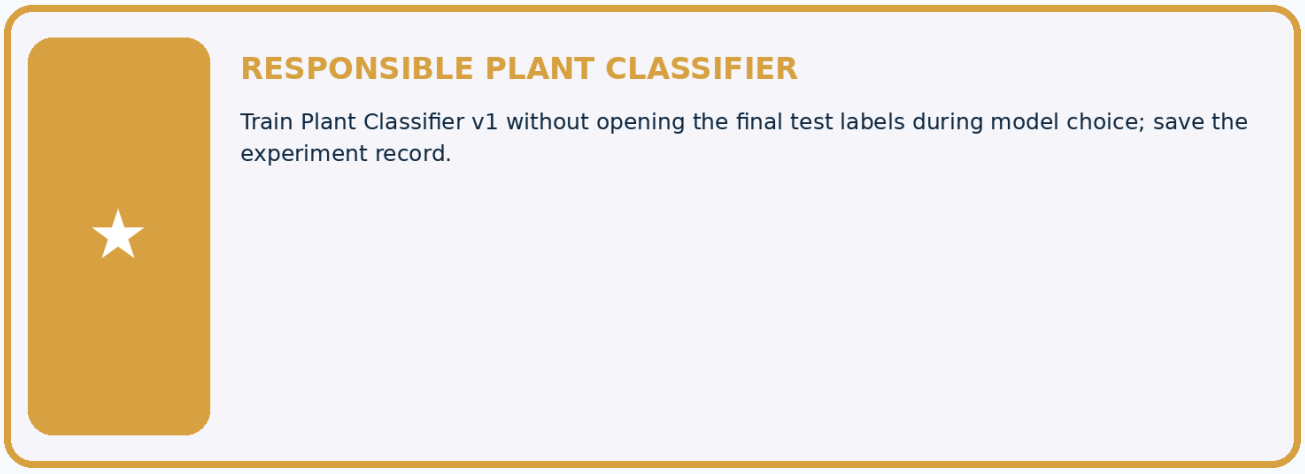
My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 17

Finish by naming fitted state, raw input and evaluation evidence

> Explain which unchanged result should reproduce even when it is wrong.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Connect **training**-only scaling and neighbour **prediction** in one **pipeline**

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
fit	learn parameters from training data	
parameter	learned numeric model value	
predict	apply fitted model to new features	
pipeline	ordered preprocessing and model steps	

Closer-explanation terms

Word	Meaning
Pipeline	An ordered sequence of data transformations and prediction steps
Fitted state	The learned values and stored examples needed to make predictions
Reproduce	Obtain the same result using the same recorded inputs and procedure

Word	Meaning
Input contract	The documented feature names order units and accepted values

▼ Extension vocabulary

Term	Meaning
Candidate rule	One possible prediction rule in a declared search
Threshold	A numerical boundary used to choose between outcomes
Training error	A mismatch between a prediction and its reference label in training data
Fitting	Using training examples to choose a model value or rule

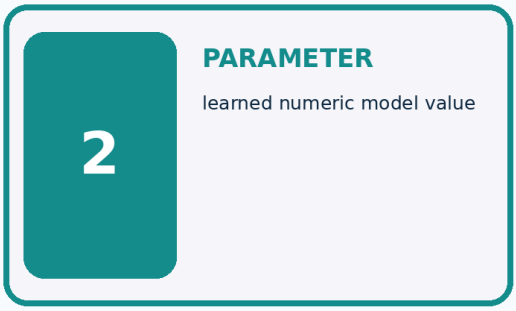
▼ Earlier wording, alternatives and picture transcripts

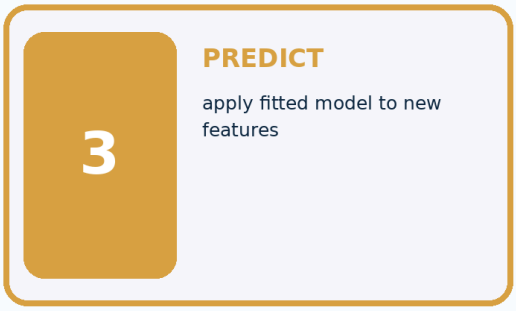
VOCABULARY

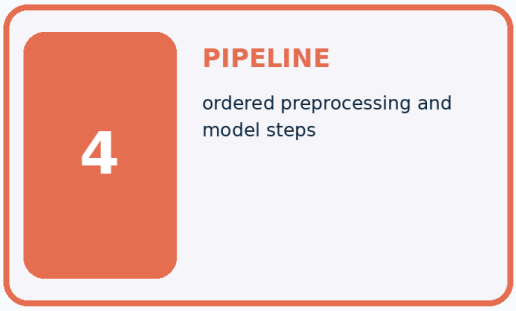
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR The model keeps learning whenever it sees test examples during evaluation.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Terms for Searching candidate rules

▼ Extra challenge · Build a stronger explanation

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: fit, **parameter**, **predict**, **pipeline**.
- CALCULATE **Reproduce** the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE The model keeps learning whenever it sees test examples during evaluation.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

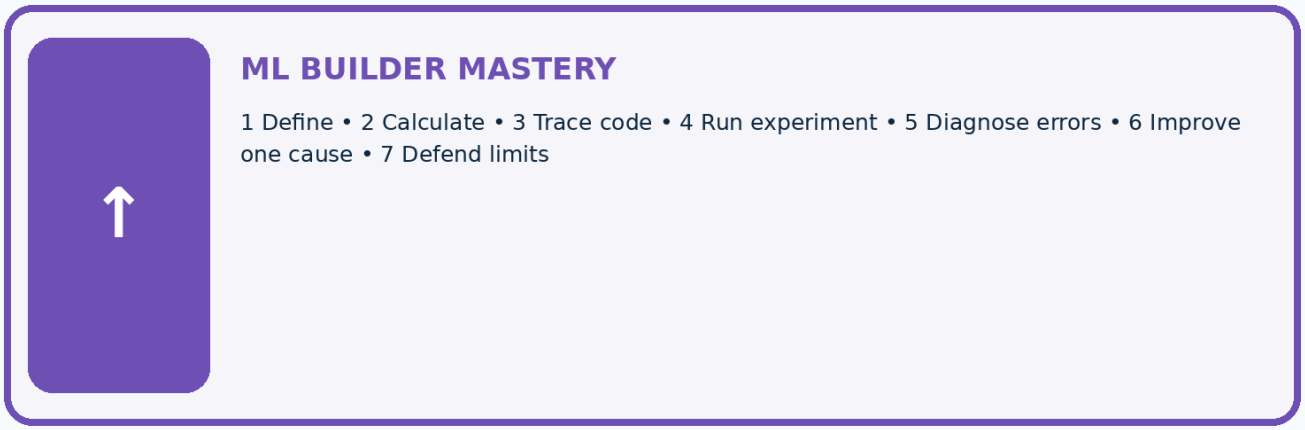
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh P/Q/R fitted-pipeline task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Run a starter classifier and record data version, split, settings, and **prediction**s.

NEXT BRIDGE Carry the experiment **evidence** into: Scores and Calibration Intuition.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Training is (0,H),(1,H),(4,S),(5,S). **Candidate rule predict**s S for spots>=3. Held-out cases are (2,H),(3,S).

Apply the rule and distinguish **fitting** from inference.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Predictions H,S give 2/2 on this tiny set. Choosing the rule is development; applying it is inference. Two cases do not establish broad performance.

Evidence or stage	Value or result
Training	0 and 1 labelled H, 4 and 5 labelled S
Supplied rule	Spots at least 3 predicts S
Test 2 spots	Predict H
Test 3 spots	Predict S

▼ Earlier wording, alternatives and picture transcripts

Week 13 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Apply a developed rule to new cases

Training is (0,H),(1,H),(4,S),(5,S). **Candidate rule predict**s S for spots>=3. Held-out cases are (2,H),(3,S).

Apply the rule and distinguish **fitting** from inference.

Predictions H,S give 2/2 on this tiny set. Choosing the rule is development; applying it is inference. Two cases do not establish broad performance.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 16

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Connect training-only scaling and neighbour prediction in one pipeline	Explain the deciding step, show a trace or test, and state one limit.
Reproduce predictions from recorded data settings and fitted values	Explain the deciding step, show a trace or test, and state one limit.
Audit a pipeline for refitting feature-order errors and evaluation leakage	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Separate a reproducible fitted **pipeline** from merely displaying **prediction**s.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 18

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Separate a reproducible fitted **pipeline** from merely displaying **prediction**s.

- Which labelled step matches this detail? Distances 30 and 10 mm select the spotted **training** record.
- 1. **Predict** neighbour
- 2. Fit transform
- 3. Transform query
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: **Predict** neighbour Distances 30 and 10 mm select the spotted **training** record. Read the picture in order. Distances 30 and 10 mm select the spotted **training** record.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Transform **training** but leave query numerically at 5. Which conclusion is supported by the supplied facts?
- 1. Distances become 15 and 55, incorrectly selecting healthy.

- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed explanation

After your attempt, compare with the supported explanation: Distances become 15 and 55, incorrectly selecting healthy. The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

Repair and keep the record

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training** accuracy replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sampling and Intended Use (ai-4/7). Goal: Define the population and conditions in an intended-use statement. Short practice: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and

the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. **Predict** first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

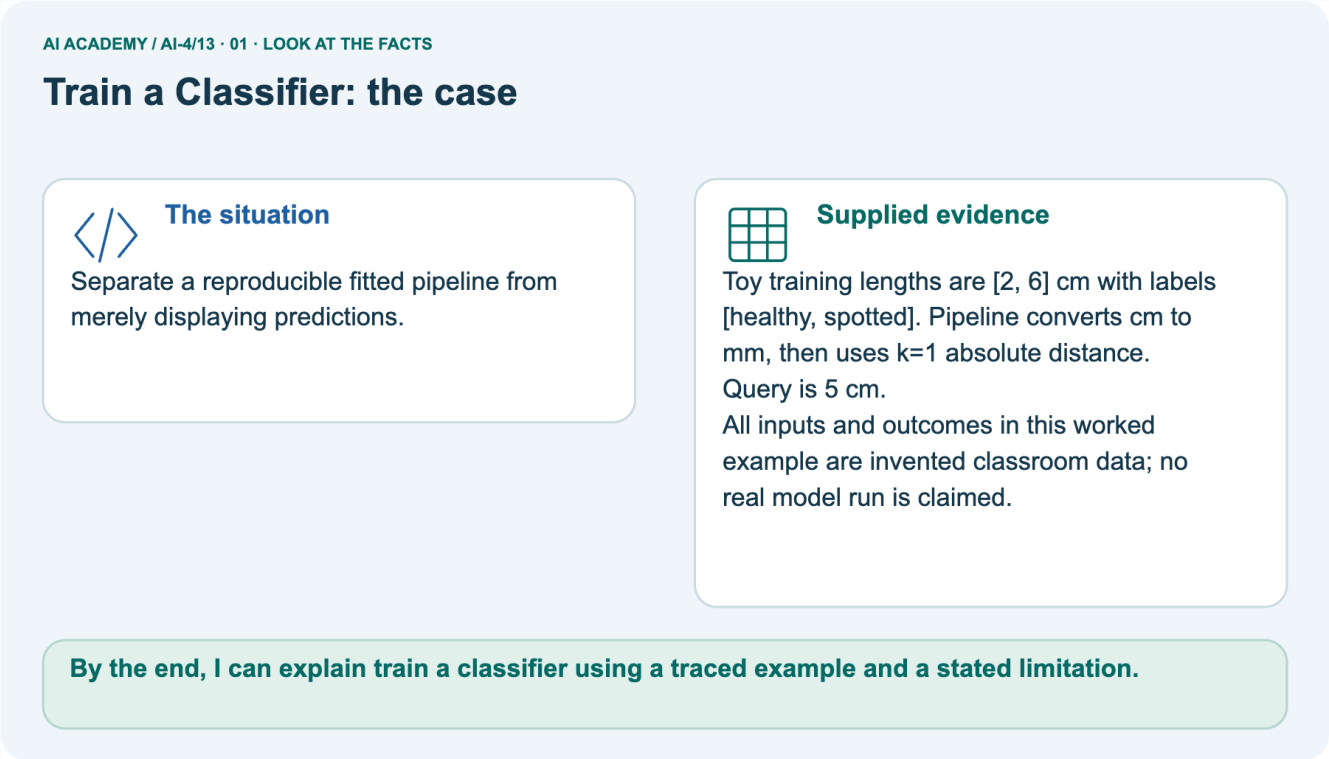
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 19

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Train a Classifier: the case



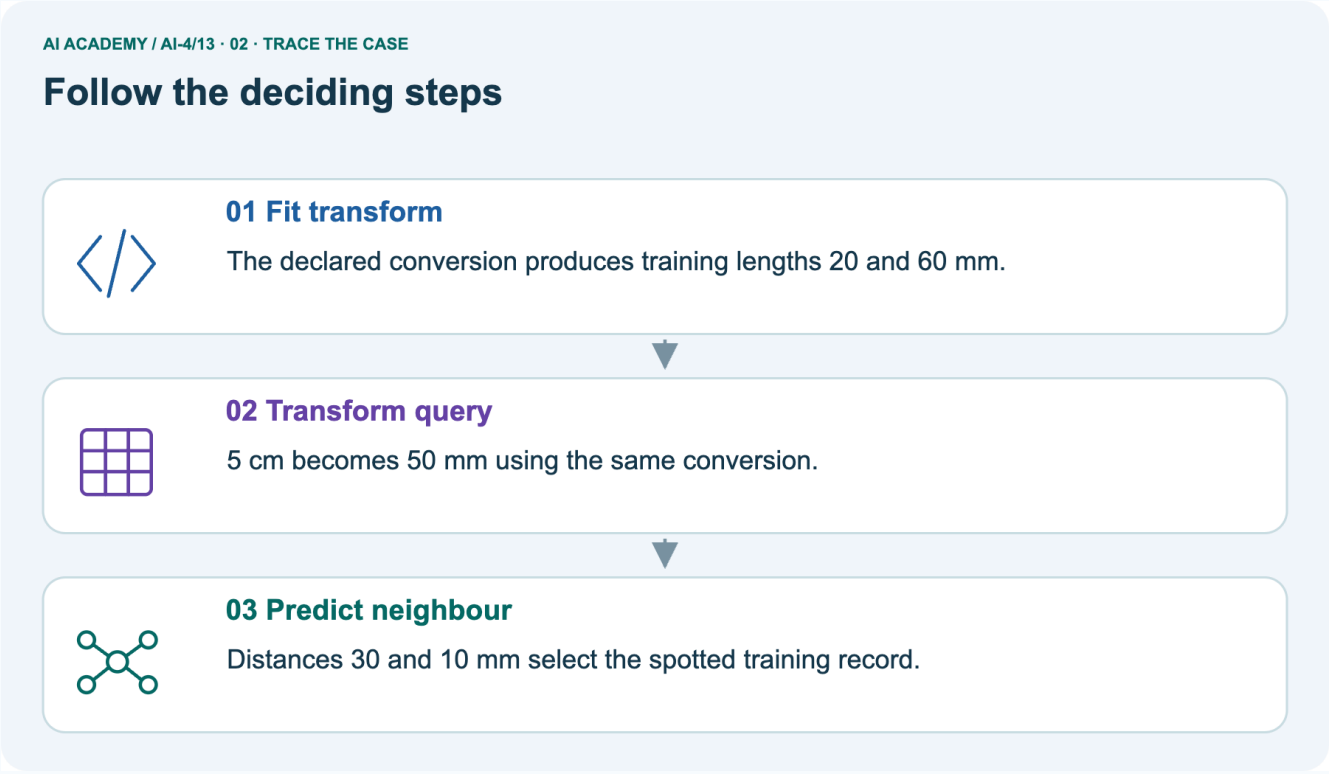
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate a reproducible fitted **pipeline** from merely displaying **prediction**s. Supplied facts: Toy **training** lengths are [2, 6] cm with labels [healthy, spotted]. **Pipeline** converts cm to mm, then uses k=1 absolute distance.

Query is 5 cm. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

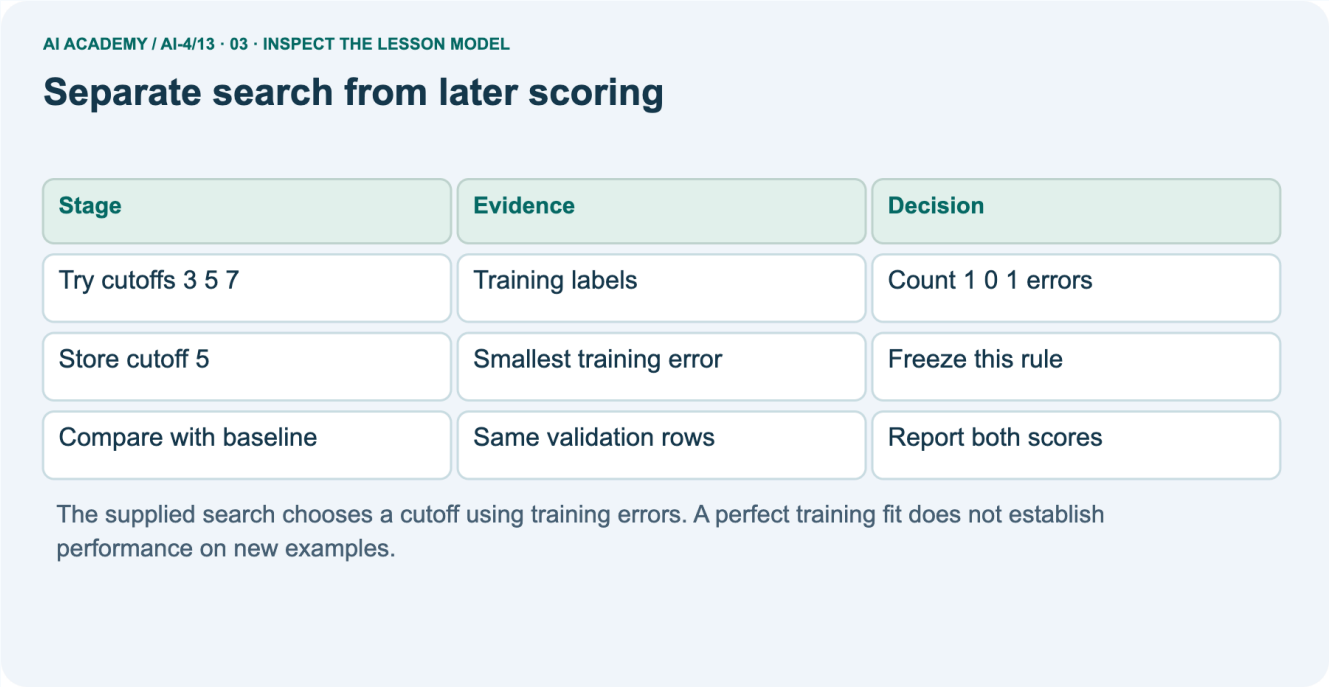


Follow the labelled picture with the matching explanation. Open the vector diagram

Fit transform: The declared conversion produces **training** lengths 20 and 60 mm. → Transform query: 5 cm becomes 50 mm using the same conversion. → **Predict** neighbour: Distances 30 and 10 mm select the spotted **training** record.

Point to each numbered stage and explain which recorded **input** it uses.

03. Separate search from later scoring

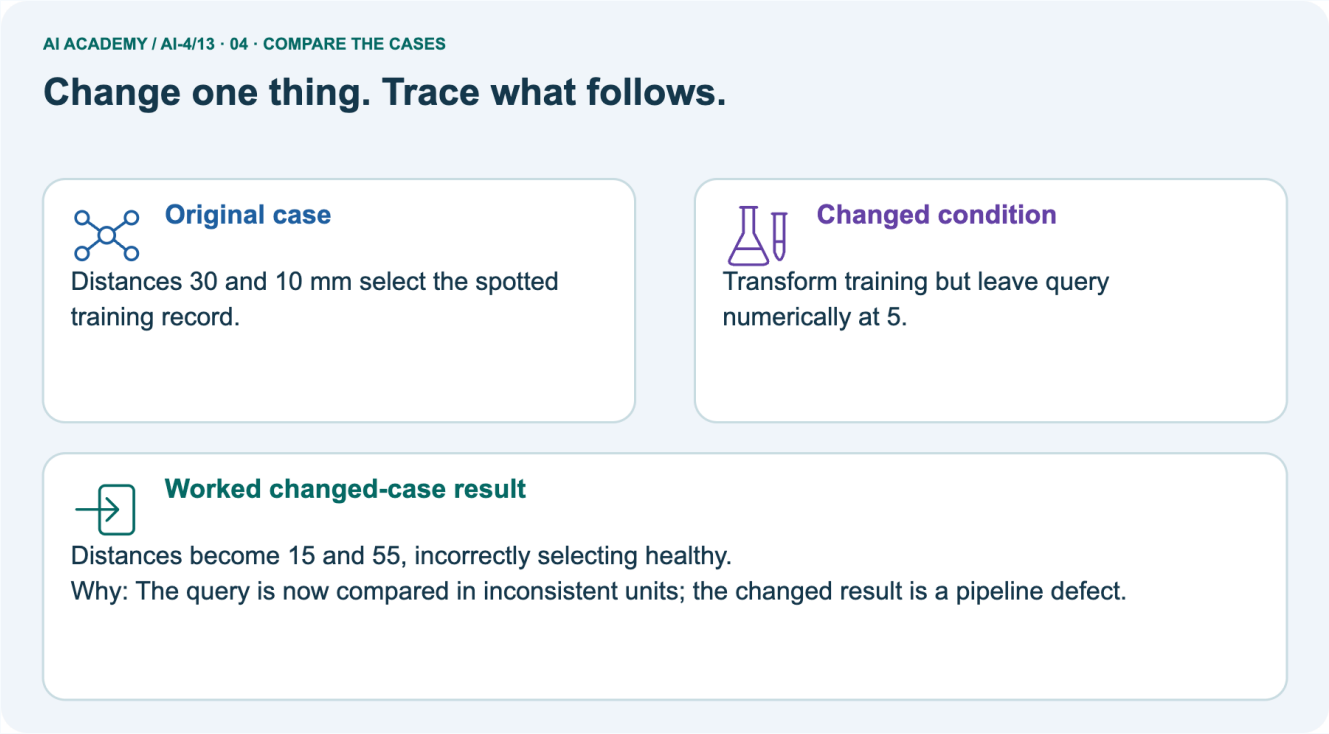


Follow the labelled picture with the matching explanation. Open the vector diagram

Separate search from later scoring. Try cutoffs 3 5 7 | **Training** labels | Count 1 0 1 errors. Store cutoff 5 | Smallest **training error** | Freeze this rule. Compare with baseline | Same validation rows | Report both scores

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



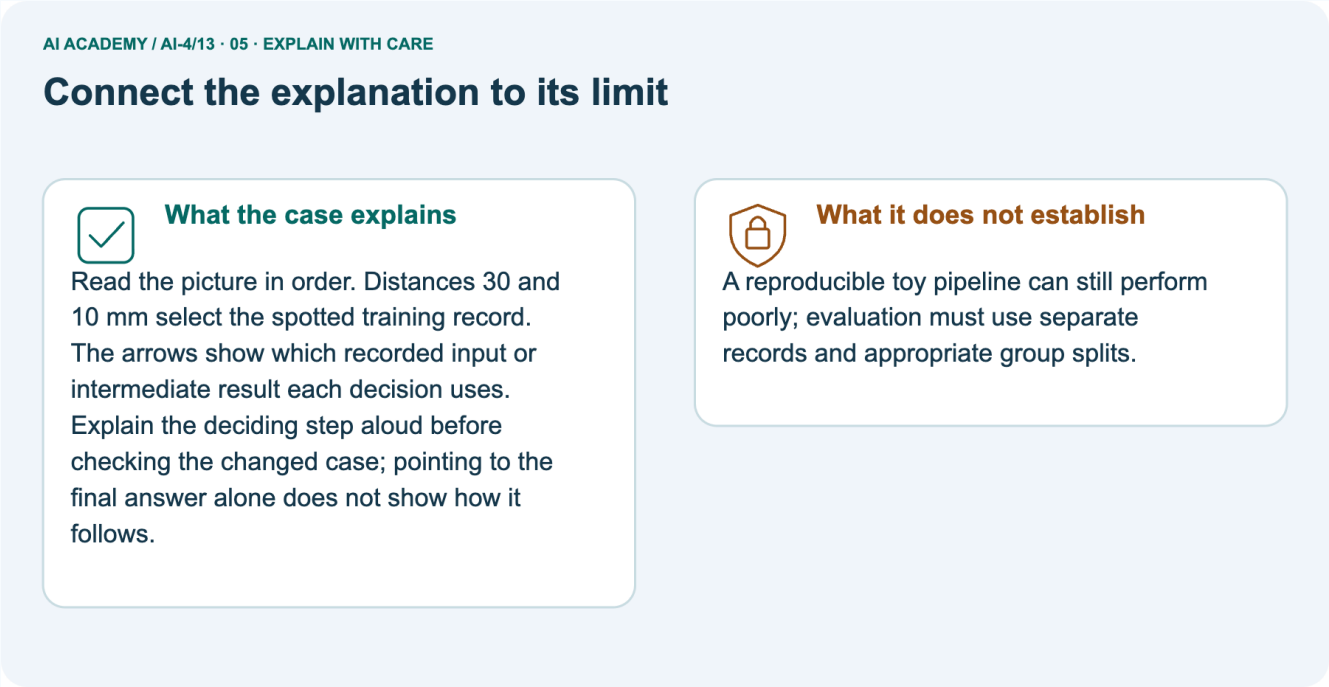
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Distances 30 and 10 mm select the spotted **training** record. Changed condition: Transform **training** but leave query numerically at 5. Expected result: Distances become 15 and 55, incorrectly selecting healthy.

Reason: The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

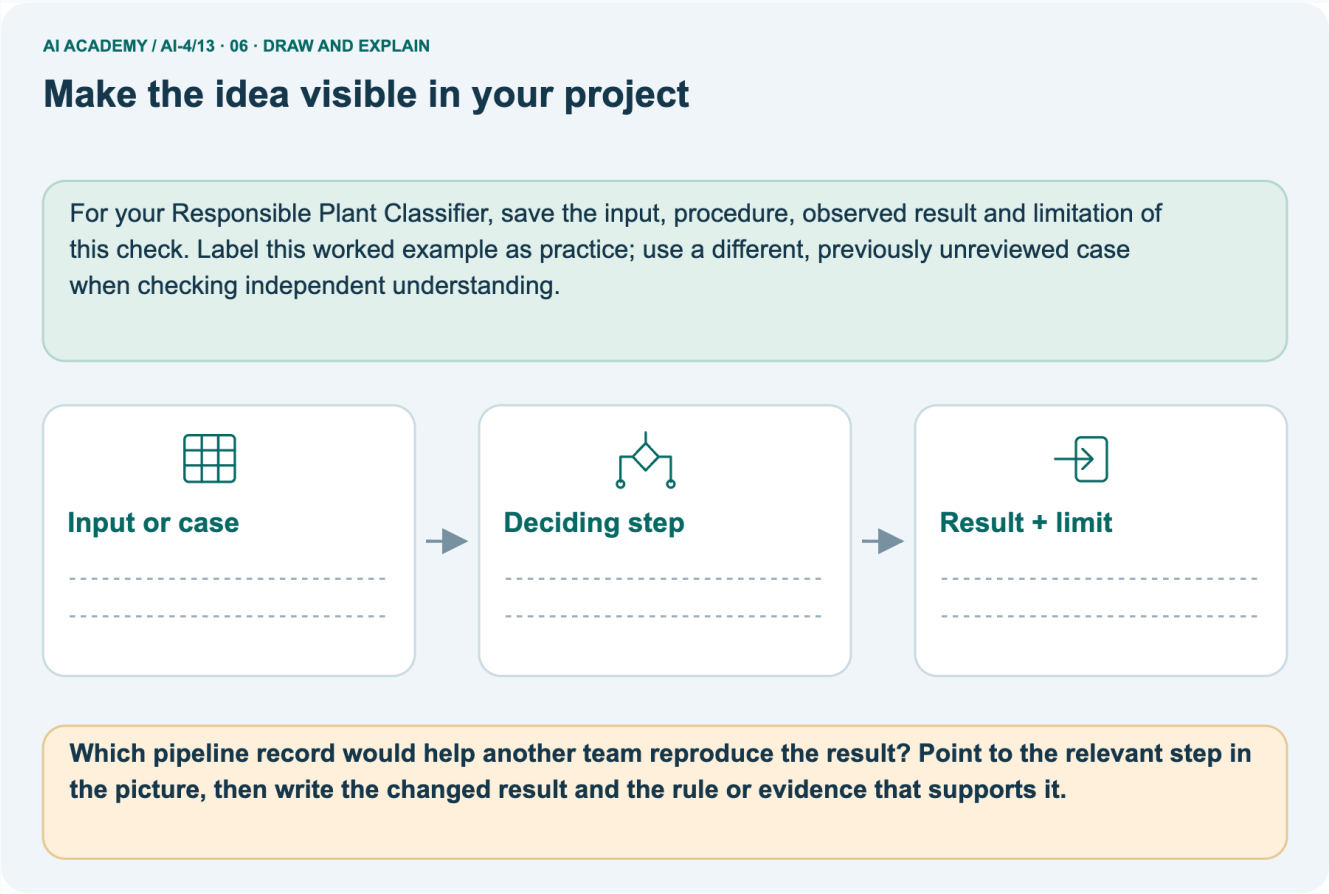
Read the picture in order. Distances 30 and 10 mm select the spotted **training** record. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A reproducible toy **pipeline** can still perform poorly; evaluation must use separate records and appropriate group splits.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Train a Classifier: the case

02. Follow the deciding steps

03. Separate search from later scoring

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional executable lab · Fit and test a threshold classifier

REFERENCE · SECTION 15

Optional executable lab · Fit and test a threshold classifier

> This retained threshold lab has its own feature/label contract; keep it separate from the main scaled-neighbour pipeline.

Complete retained lab and evidence

Fit and test a threshold classifier

The program chooses a **threshold** using **training** labels only. Spotted is 1 and healthy is 0. Candidate **threshold**s lie between consecutive **training** values. Test rows are used after selection.

```
"""Fit a threshold using training data, then evaluate untouched examples."""
train = [(0, 0), (1, 0), (4, 1), (5, 1)]
values = sorted({x for x, y in train})
if len(values) < 2:
    raise ValueError("Two distinct training values are required")
candidates = [(a + b) / 2 for a, b in zip(values, values[1:])]
```

```
def errors(threshold):
    return sum(int(x >= threshold) != y for x, y in train)


threshold = min(candidates, key=errors)
test = [(2, 0), (3, 1)]
predictions = [int(x >= threshold) for x, y in test]
correct = sum(pred == y for pred, (x, y) in zip(predictions, test))
print(threshold, predictions, correct, len(test))
```

Expected **output**: 2.5 [0, 1] 2 2

Candidate **threshold**s 0.5, 2.5 and 4.5 have **training error**s 1, 0 and 1. The selected **threshold** is 2.5. Both test **prediction**s are correct.

This tiny artificial example demonstrates **fitting** and evaluation; it does not establish reliability on real leaf photos.

Independent code change

Change the final test row to (3, 0), keeping **training** unchanged. **Predict** before running.

My **prediction** and **evidence**

> **Write or draw your reasoning and evidence.**

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.